

Week 1 – Monday: Orientation, Setup & Cybersecurity Careers

Cybersecurity Course | Fully Remote



Course Overview

- Welcome to Cybersecurity!
- - Combination of lectures and hands-on labs
- - Focus: Practical skills in security, scripting, and systems
- - Tools: VirtualBox, Kali Linux, Windows, Python, Bash

Orientation Goals

- - Understand course structure and expectations
- - Set up communication tools (email, Slack, Jira)
- - Install VirtualBox
- - Import Kali Linux VM
- - Take your first snapshot
- - Explore your learning environment

What is Cybersecurity?

- Cybersecurity involves protecting systems, networks, and data from digital attacks.
- Key Concepts:
 - - Confidentiality, Integrity, Availability (CIA Triad)
 - - Threats, vulnerabilities, and risks
 - - Common terms: firewall, encryption, phishing

Cybersecurity Careers

- - Security Analyst
 - - Penetration Tester
 - - Network Engineer
 - - SOC Analyst
 - - Risk Manager
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- Certifications: CompTIA Security+, CEH, CISSP, AWS CCP

Virtualization Setup

- Download and install VirtualBox
- Import VM using File > Import Appliance
- Start the VM and log in (kali/kali)
- Take a snapshot for backup
- Explore system settings & interface
- https://drive.google.com/file/d/1WKQg4CI59xN9BTsuPOMMfWw_oCbI92Ya/view

FINALIZING VM SETUP

- Take a snapshot in VirtualBox
- Adjust VM settings:
 - Shared Clipboard: Enable bidirectional
 - Network: Set to NAT or Bridged
- Kali Linux internet access
 - (ping google.com)

INTERACTION BETWEEN HOST AND VM

- VMs are like a sandbox: Safe testing environment
- Host-VM interaction:
 - Shared Clipboard: Copy-paste between systems.
 - Networking: NAT (private) vs. Bridged (network device).

Why it matters: Test attacks without risking your real system

Course Expectations

- Show up on time
 - 6:30 P.M CST
 - Monday-Thursday
- Participate
- Ask questions
- Cameras on

Q&A